Software Requirements Specification

ID	Type	Description	Source	Rationale	Priority	Status	Ver. Method
SRS-1	Information	1 Introduction	IEEE		Low	Draft	Demonstration
SRS-2	Information	1.1 Purpose Identity theft and phishing are the common dangers of online shopping. Secure Socket Layer (SSL) encryption prevents the interception of consumer information in transit between the consumer and the online merchant. However, one must still trust merchant and its employees not to use consumer information for their own purchases and not to sell the information to others. We hereby propose a new method, that uses steganography (text based) and visual cryptography, which minimizes information sharing between consumer and online merchant but enable successful fund transfer from consumer's account to merchant's account thereby safeguarding consumer information and preventing misuse of information at merchant side.	IEEE		Low	Draft	Demonstration
SRS-3	Information	1.2 Scope 1. **Product Name –** ONLINE PAYMENT SYSTEM USING STEGANOGRAPHY AND VISUAL CRYPTOGRAPHY. 1. **Purpose** – Improve the existing E-Commerce & online shopping retailer's application security 1. **Objective** – Using steganography (text based) and visual cryptography, which minimizes information sharing between consumer and online merchant but enable successful fund transfer from consumer's account to merchant's account thereby safeguarding consumer information and preventing misuse of information at merchant side. The method proposed is specifically for E-Commerce but can easily be extended for online as well as physical banking 4. **Goals** – 1. Providing limited information which is necessary for fund transfer during online shopping 2. Safeguarding customer data and increasing customer confidence and preventing identity theft 3. Maintain reusability while designing the software 4. Convince the government agencies to use our software.	IEEE		Medium	Draft	Demonstration

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SRS-4	Information	1.3 Product perspective A rapid growth in E-Commerce market is seen in recent time throughout the world. With ever increasing popularity of online shopping, Debit or Credit card fraud and personal information security are major concerns for customers, merchants and banks specifically in the case of CNP (Card Not Present). This paper presents a new approach for providing limited information only that is necessary for fund transfer during online shopping thereby safeguarding customer data and increasing customer confidence and preventing identity theft. The method uses combined application of steganography and visual cryptography for this purpose.	IEEE		Low	Draft	Demonstration
SRS-5	Interface Requirement	1.3.1 System interfaces Web Application runs on various operating systems supporting modules of various frameworks.	IEEE		Low	Draft	Demonstration
SRS-6	Interface Requirement	1.3.2 User interfaces Steps Involved in the Online Shopping of Consumer: • Creation of Shares • Registration with the merchants' site • Selecting item on merchant's website • Redirection to payment gateway (CA) • Insertion of share key • Successful Payment Message to consumer	IEEE		Medium	Draft	Demonstration
SRS-7	Interface Requirement	1.3.3 Hardware interfaces Supported Devices – Personal Computers, Laptops, Digital Platform where online shopping takes place often	IEEE		Medium	Draft	Demonstration
SRS-8	Interface Requirement	1.3.4 Software interfaces • Operating Systems – Ubuntu 18.04 and higher • Database Used – MySQL • Modules Required – Flask's MySQLdb, Opencv or PIL, AES, SHA • Web Based Functionality co-ordinated with Python 3 using Flask Framework	IEEE		High	Draft	Demonstration
SRS-9	Interface Requirement	1.3.5 Communications interfaces • High Level Use of HTTP protocol • Socket Size (Bandwidth) for 8192 Bytes for receiving and sending data (minimum requirement)	IEEE		High	Draft	Demonstration
SRS-10	Non-func. Requirement	1.3.6 Memory constraints Web application needs some amount of database to store shares of particular consumer. So memory allocated must be dynamic and flexible to grow	IEEE		High	Draft	Demonstration

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SRS-11	Information	1.3.7 Operations 1. This software is designed to work on Ubuntu operating system with version 18.04 or higher. 2. The software is developed in python 3 language. 3. The software doesn't interfere with any other running process	IEEE		Medium	Draft	Demonstration
SRS-12	Section	1.3.8 Site adaptation requirements TBD	IEEE		Low	Draft	Demonstration
SRS-13	Section	1.3.9 Interfaces with services CA like Payment Gateways	IEEE		Low	Draft	Demonstration
SRS-14	Functional Requirement	1.4 Product functions • Steganography using ASCII code:- Hiding message using Text Steganography Technique 1. Input Encoding --> Encoding secret message with cover text 2. Decoding --> Decoding/recovering the secret message • Visual Cryptography :- Encrypting and Decrypting secret message image 1. Encryption --> Dividing secret message into sufficient no. of shares 2. Decryption --> stacking all the shares to retrieve secret image	IEEE		High	Draft	Demonstration
SRS-15	Non-func. Requirement	1.5 User characteristics • Consumers -- Who want to keep their data safe can enforce this methodology on merchants in online shopping industry. • Merchants -- Here the Customers are merchants who want to take consumers into confidence of providing security of data. • Banks – Governing the addressed system of dual encryption • Middle Authorities – like payment gateways • Government of a country or state which can ask for merchants and consumers support for using this methodology.	IEEE		High	Draft	Demonstration

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SRS-16	Information	1.6 Limitations • Size of text generated after encryption using steganography may result into space consumption but it gives better standards of encryption. • It is the responsibility of consumer to keep the shares of encrypted image safe.	IEEE		Low	Draft	Demonstration
SRS-17	Non-func. Requirement	1.7 Assumptions and dependencies 1. Required Modules (will be identified in the later versions) are to be installed inside the system. 2. Hardware of user's PC alongside user's default browser are compatible with AES 256, SHA 256.3) 3. User has Ubuntu 18.04 or higher operating system. 4. User has python3 installed.	IEEE		Low	Draft	Demonstration
SRS-18	Section	1.8 Definitions • Visual Cryptography- Cryptographic technique which allows visual information to be encrypted in such a way that decrypted information appears as visual image. • Encryption – Hiding the information using cryptography • Decryption – Retrieving the original information back. • Cryptography – The art of writing or solving codes. • AES – Advanced encryption standard	IEEE		Low	Draft	Demonstration
SRS-19	Section	1.9 Acronyms and abbreviations • VC -- Visual Cryptography • CA -- Certified Authority • AES – Advanced Encryption Standards	IEEE		Low	Draft	Demonstration
SRS-20	Section	2 Requirements	IEEE		Medium	Draft	Demonstration
SRS-21	Interface Requirement	2.1 External interfaces • Operating Systems – Ubuntu 18.04 and higher • Database Used – MySQL • Modules Required – Flask's MySQLdb, Opencv or PIL, AES, SHA • Web Based Functionality co-ordinated with Python 3 using Flask Framework	IEEE		High	Draft	Demonstration

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SRS-22	Functional Requirement	2.2 Functions **Encryption** • Description and Priority 1. This Feature is used by both steganography and visual cryptography. 2. Secret key is first to be converted into unique code using Vedic codes and visual cryptography converts image into shares. • Stimulus/Response Sequences : 1. Registration of user which generates shares 2. Right choice of shares to be done while making payment **Decryption** • Description and Priority 1. This feature involves decryption algorithm using visual cryptography. 2. Once all the shares are combined to make one single image and then it is decrypted we get a normal image 3. Which is then is to be decoded to get secret key. 4. This secret key is then sent to bank. • Stimulus/Response Sequences: 1. Search for particular share in database. 2. Combining it with consumers' Shares. 3. Sending the key to Bank Verification	IEEE		High	Draft	Demonstration
SRS-23	Functional Requirement	2.3 Usability requirements • The product must provide an user friendly bridge between merchants' website and bank database. • The number of physical interactions should be minimum.	IEEE		Low	Draft	Demonstration
SRS-24	Functional Requirement	2.4 Performance requirements • Availability of Proper Bandwidth. • The software should use minimum memory by optimizing the algorithms for space complexity.	IEEE		Low	Draft	Demonstration
SRS-25	Functional Requirement	2.5 Logical database requirements TBD	IEEE		Low	Draft	Demonstration

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SRS-26	Non-func. Requirement	2.6 Design constraints 1. During payment, merchant's payment system requires to direct the shopper to CA's portal but fraudulent merchant may direct shopper to a portal similar to CA's portal but of its own making and get hold of customer own share. 2. Need to carry the secret image share to authenticate the payment process 3. Use of text steganography involves encoding using Vedic Codes which is indeed described for alphabets. In this case, the key which will be used by the consumer to generate shares must contain alphabets.	IEEE		High	Draft	Demonstration
SRS-27	Information	2.7 Standards compliance Rules and Regulations suggested by SSL certificates and Certified Authorities	IEEE		Low	Draft	Demonstration
SRS-28	Section	2.8 Software system attributes • With increasing number of customers, Database must be flexible to grow in size. • The software will be robust, specific to particular consumer and verifiable where every entry of data in the database will be verified	IEEE			Draft	Demonstration
SRS-29	Section	3 Verification • HTML Pages generating interface to consumer • Certified Authority Role which will be performed by this product • Database for certified Authority • Transaction involving interface provided by Bank	IEEE		Low	Draft	Demonstration
SRS-30	Section	4 Supporting information **End Users:** • Banks: Checking the authentication and processing fund transfer to merchant • CA: Storing shares and decrypting secret image using decryption algorithm. • Consumers: Generating shares while registration and using them as secret key while payments. • Merchants: Providing Interface to build a bridge between consumer and CA	IEEE		Medium	Draft	Demonstration
SRS-31	Section	5 References • Souvik Roy and P. Venkateswaran, "Steganography and Visual Cryptography: Applications", 2014 IEEE Students' Conference on Electrical, Electronics and Computer Science. • Mukta Sharma, Neha Sabharawal, " A Glance at Cryptographic Techniques used in Indian Banks". • Robert S. Pressman, 'A Practitioner's Approach'	IEEE		Medium	Draft	Demonstration

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